QP Code: 25901275

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEFTQ201 - FOOD ENGINEERING

(2024 ADMISSION ONWARDS)

Duration: 2 hours Maximum Marks: 70

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I)and Appreciation(Ap)

Part A

Short Answer type

Answer any 10 questions

Each question carries 2 marks

1. Define thermal diffusivity. [K] / [CO2]

Thermal diffusivity is the measure of how rapidly heat is transferred through a material. It depends on thermal conductivity, density, and specific heat capacity. It is expressed as $K = k / (\rho Cp)$.

2. Define fundamental units. [K] / [CO1]

Fundamental units are basic physical units that are independent and cannot be derived from other units. Examples include meter, kilogram, second, and kelvin.

3. How to convert kelvin to celsius scale? [A] / [CO1]

The temperature in Celsius is obtained by subtracting 273.15 from the Kelvin value. The relation is $°C = K - 273.15$.

4. Viscosity is an important property in food processing operations. Justify. [E] / [CO4]

Viscosity represents the resistance of a fluid to flow. It plays a critical role in food processing as it affects pumping, mixing, heat transfer, and texture. High viscosity materials require more energy for processing and influence product quality.

5. Write mass balance equation for steady state system. [U] / [CO1]

In a steady state system, there is no accumulation of mass within the system. Therefore, the mass entering the system is equal to the mass leaving the system, expressed as Input = Output.

6. Differentiate between coarse crushing and fine grinding. [U] / [CO5]

Coarse crushing reduces materials into relatively larger fragments, whereas fine grinding produces very small and uniform particles with increased surface area. Grinding requires higher energy compared to crushing.

7. What happens to water during freezing? [U] / [CO3]

During freezing, water is converted into ice crystals. This process leads to expansion in volume and reduction in water activity. Large ice crystals may damage cellular structures in food.

8. Define ohmic heating. [U] / [CO2]

Ohmic heating is a method in which electric current is passed through food, causing internal heat generation due to electrical resistance. This results in rapid and uniform heating.

9. Comment on lyophilization. [K] / [CO5]

Lyophilization, also known as freeze drying, is a process where water is removed from frozen material by sublimation under vacuum. It helps in preserving nutritional quality, flavor, and structure.

10. Define rheology. [K] / [CO4]

Rheology is the study of the flow and deformation of materials, particularly fluids and semi solids. It is important in understanding the behavior of food products during processing.

11. State the methods to reduce water activity in foods. [K] / [CO3]

Water activity in foods can be reduced by drying, addition of solutes such as salt or sugar, freezing, and concentration. These methods inhibit microbial growth and enhance preservation.

12. What is the condition for an adiabatic process? [U] / [CO3]

An adiabatic process is one in which there is no heat exchange between the system and its surroundings. The condition is that heat transfer (Q) is equal to zero.

Part B

Short Essay Questions

Answer any 6 questions

Each question carries 5 marks

13. Describe the construction and working of a scraped surface heat exchanger. [U] / [CO2]

A scraped surface heat exchanger consists of a cylindrical shell with a rotating shaft fitted with blades. The blades continuously scrape the inner surface to prevent fouling and ensure efficient heat transfer. It is widely used for viscous and heat sensitive food products.

14. Describe surface tension and explain its importance in food processing operations. [U] / [CO4]

Surface tension is the force acting at the surface of a liquid due to cohesive forces between molecules. It plays a significant role in emulsification, foaming, droplet formation, and stability of food systems.

15. Write a short note on equipments used for size reduction of liquid foods. [A] / [CO5]

Equipments such as homogenizers and colloid mills are used for size reduction of liquid foods. These devices reduce particle size, improve texture, and enhance stability of emulsions.

16. Explain rotary pumps. [U] / [CO4]

Rotary pumps are positive displacement pumps that use rotating components to move fluids. They are suitable for handling viscous and shear sensitive food materials and provide a steady flow.

17. Differentiate First and Second Laws of Thermodynamics. [An] / [CO3]

The first law states that energy cannot be created or destroyed but can be transformed from one form to another. The second law states that heat flows from a higher temperature to a lower temperature and that entropy of an isolated system increases.

18. Write a short note on emissivity and its role in radiation heat transfer. [An] / [CO2]

Emissivity is the ability of a surface to emit thermal radiation compared to a perfect black body. It influences the efficiency of heat transfer in processes such as baking and drying.

19. Explain U-tube manometer. [U] / [CO4]

A U-tube manometer is a device used to measure pressure differences using a liquid column. The difference in liquid levels in the two arms indicates the pressure difference.

20. Compare pasteurization and sterilization based on thermodynamic principles. [An] / [CO3]

Pasteurization involves heating food to moderate temperatures to destroy pathogenic microorganisms, whereas sterilization involves higher temperatures to eliminate all microorganisms including spores. Sterilization requires greater energy input.

Part C

Long Essay Questions

Answer any 2 questions

Each question carries 10 marks

21. Explain basic drying process with drying rate curve and stages. [An] / [CO5]

Drying is the process of removing moisture from food using heat. The drying rate curve consists of three stages: initial constant rate period, falling rate period, and final equilibrium stage. Moisture removal initially occurs rapidly and then decreases as bound water remains.

22. Discuss the concepts of internal energy and enthalpy and explain their significance in food processing operations. [An] / [CO3]

Internal energy is the total energy contained within a system, including kinetic and potential energy of molecules. Enthalpy is the total heat content of a system and is given by H = U + PV. These concepts are important in designing heating and cooling processes.

23. Compare natural convection and forced convection with respect to mechanism, heat transfer coefficient, and applications in food processing. [U] / [CO2]

Natural convection occurs due to density differences caused by temperature variation, whereas forced convection involves external forces such as pumps or fans. Forced convection has a higher heat transfer coefficient and is widely used in industrial processes.

24. Derive the continuity equation. [C] / [CO4]

The continuity equation is derived from the principle of conservation of mass. For a steady flow system, the mass flow rate remains constant, expressed as $\rho A_1 V_1 = \rho A_2 V_2$. It is used in fluid flow analysis.